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Input device

Abstract

An input device includes a control surface for a color correction system. The control surface includes a housing with an upwardly facing control panel. The control panel has a proximal edge which is nearest a user and a distal edge that is furthest from a user in normal use. The control panel includes a plurality of controls. The plurality of controls includes: a plurality of trackballs, wherein each trackball comprises a ball and a control ring, said ball cooperating with at least one encoder to generate a multi-dimensional control signal based on motion of the ball, and said control ring cooperating with at least one encoder to generate a one dimensional control signal based on the rotational motion of the ring, wherein the ball of said trackball is mounted concentrically with said control ring; a plurality of control buttons; and a plurality of knobs coupled to respective rotary encoders.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6282082	12/2000	Armitage et al.	N/A	N/A
7441193	12/2007	Wild	345/173	G06F 3/0219
8363014	12/2012	Leung et al.	N/A	N/A
10198045	12/2018	Kong	N/A	H04B 5/26
2005/0231476	12/2004	Armstrong	345/161	G06F 3/0346
2008/0048984	12/2007	Ikeda	N/A	N/A
2017/0262080	12/2016	Armstrong	N/A	G06F 3/0346
2018/0210506	12/2017	Hui	N/A	G06F 1/1654
2019/0138121	12/2018	Selby	N/A	G06F 3/03549

OTHER PUBLICATIONS

Filmlight, “Blackboard 2”, [retrieved from internet on Jun. 16, 2023] <URL:https://web.archive.org/web/20220523133850/https://www.filmlight.ltd.uk/pdf/datasheets/FL-BL-DS-0794-Blackboard2.pdf> published on May 23, 2022 as per Wayback Machine, 2 pages. (Year: 2022). cited by examiner

Blackmagic Design, “DaVinci Resolve Advanced Panel” [retrieved from internet on Jun. 19, 2023] < URL: https://web.archive.org/web/20220123034908/https://www.blackmagicdesign.com/api/print/to-pdf/products/davinciresolve/techspecs/W-DRE-06?filename=davinci-resolve-advanced-panel-techspecs.pdf > published on Jan. 23, 2022 as per Wayback Machine, 4 pages. cited by applicant

Filmlight, “Blackboard 2”, [retrieved from internet on Jun. 16, 2023]< URL: https://web.archive.org/web/20220523133850/https://www.filmlight.ltd.uk/pdf/datasheets/FL-BL-DS-0784-Blackboard2.pdf > published on May 23, 2022 as per Wayback Machine, 2 pages. cited by applicant

Australian Government, IP Australia, International-type search report for AU Application No. 2022902565, dated Jun. 21, 2023, 21 pages. cited by applicant

Avid Technology, Inc. “AvidDock Guide, Installation and Overview”, © 2021, https://resources.avid.com/SupportFiles/ProMixing/Avid_Dock_Guide_v2021.6.pdf, 39 pages. cited by applicant
Thornton, M., posted Mar. 25, 18 “The History of Pro Tools—2012 to 2018”, https://www.production-expert.com/home-page/2018/3/8/the-history-of-pro-tools-2012-to-2018, including section “2015—Avid Pro Tools Dock Control Surface”, downloaded Sep. 5, 2023. cited by applicant
Extended European Search Report for EP Application No. 23195805.9, dated Nov. 28, 2023, 10 pages. cited by applicant
Carman Robbie “Focus On The New Blackmagic Resolve Mini Panel”, Apr. 9, 2017, pp. 1-12, XP093103655, Retrieved from the Internet: URL:https://web.archive.org/web/20220630195606/https://mixinglight.com/color-grading-tutorials/color-correction-gear-headmarch-2017-edition/ [retrieved on Nov. 20, 2023]. cited by applicant
Elwyn, Jonny “Affordable Colour Grading Control Surfaces”, Aug. 17, 2022, pp. 1-42, XP093103491, Retrieved from the Internet: URL:https://web.archive.org/web/20220817101452/https://jonnyelwyn.co.uk/film-video-editing-tools-for-editors/affordablecolour-grading-control-surfaces/ [retrieved on Nov. 20, 2023]. cited by applicant
Avid Technology, Inc. “Avid Dock”, Nov. 20, 2023, pp. 1-3, XP093103633, http://www.avid.com/products/avid-dock Retrieved from the Internet: URL:https://web.archive.org/web/20220808072140/https://www.avid.com/products/avid-dock [retrieved on Nov. 20, 2023]. cited by applicant

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Background/Summary

BACKGROUND
Technical Field
(1) The present disclosure relates to an input device.
(2) The input device takes the form of a control surface for a color correction system.
BRIEF SUMMARY
(3) In the context of the present disclosure, a color correction system will typically include a color correction application running on a computer system. The color correction application could be a standalone color correction application or an application such as a non-linear editor that has a color correction capability as a subset of its functionality. One such example of such a color correction application is Davinci Resolve® from Blackmagic Design. In some embodiments the color correction system may include a special purpose computer system (e.g., video processing appliance, video card or the like) configured to perform color correction of video or still images. The color correction system will include a display device including a screen on which a user may view a graphical user interface for the color correction system and images or videos that are subject to the color correction.
(4) The term “color correction” as used herein (including use in adjectival or nominal form) is intended to have a meaning that includes “color grading” and color adjustment that may be performed on digital video and images for other purposes.

(5) Embodiments of the present disclosure are adapted for use with color correction systems that include a portable display device. In a preferred embodiment, the present disclosure is adapted for use with a color correction system comprising a color correction application running on, or operable via a tablet computer, smart phone or other device of a similar form factor.
(6) The systems, devices, methods and approaches described in this specification, and components thereof are known to the inventors. Therefore, unless otherwise indicated, it should not be assumed that any of such systems, devices, methods, approaches or their components described are citable as prior art merely by virtue of their inclusion in this section, or that such systems, devices, methods, approaches and components would ordinarily be known to a person of ordinary skill in the art.
(7) In a first aspect there is provided a control surface for controlling a color correction system. The control surface may include a housing including an upwardly facing control panel, said control panel having a proximal edge which is nearest a user in normal use and a distal edge that is furthest from a user in normal use.
(8) The control panel includes a plurality of controls that may include: a plurality of trackballs, wherein each trackball comprises a ball and a control ring, said ball cooperating with at least one encoder to generate a multi-dimensional control signal based on motion of the ball, said control ring cooperating with at least one encoder to generate a one dimensional control signal based on the rotational motion of the ring, wherein the ball of said trackball is mounted concentrically with said control ring; a plurality of control buttons; a plurality of knobs coupled to respective rotary encoders;
(9) The control surface can also include: a display stand adjacent the distal edge of the control panel for supporting a display device of the color correction system in an upright position in use; a power storage system contained within the housing, said power storage system being arranged to supply power to the control surface in use; a wireless communications interface over which control signal from the plurality of controls are transmitted for use in controlling the color correction system.
(10) In some embodiments, the power supply system may include one or more of: a battery; and a charging system.
(11) In some embodiments, the wireless communications interface operates according to one of the following wireless communications methodologies: an IEEE 802.11 wireless standard; Bluetooth, ZigBee or other IEEE 802.15 standard, free-space optical communication.
(12) The display stand may comprise any one or more of the following: boss, protrusion, rib, lip, flange, beam, channel, step, hole, recess, aperture, indentation, slot, cushion, gripping surface, frame, wall, ledge, shelf, frame, hook and cradle.
(13) Such elements of the display stand can be arranged to perform any one or more of the following functions: support the weight of the display device of the color correction system in use; constrain any one or more of sliding, twisting, or tilting of the display device of the color correction system in use; act as a fulcrum that supports display device of the color correction system in use; grip the display device of the color correction system in use.
(14) In some embodiments, the display stand may comprise a channel extending adjacent to the distal edge of the control panel, said channel being adapted to receive an edge of a display device of the color correction system in use, said channel having a front support surface that limits movement of a supported display device of the color correction system in a proximal direction, and a rear support surface that constrains tilting of the supported display device of the color correction system in a distal direction.
(15) Supporting the display device of the color correction system in an upright position may comprise supporting the display device of the color correction system such that it is tilted in a distal direction at an angle of more than 10 degrees from vertical.
(16) The display stand may include an affordance to enable a user to touch a screen of a supported

display device of a color correction system, adjacent to the lowermost edge of the screen. The affordance may enable a user to touch a central part of the lowermost edge of the screen. The affordance may be a recess, indentation, groove, or gap.
(17) In some embodiments, the arrangement of trackballs is symmetrical about a centerline of the control panel.
(18) In some embodiments, the display stand can support a display device of the color correction system in a position such that a screen of the display device is symmetrical about said centerline. The display stand may be symmetrical about said centerline.
(19) In some embodiments, the buttons on the control panel may be arranged in groups, and wherein said groups of buttons are positioned symmetrically about the centerline of the control panel. An arrangement of buttons within at least one pair of symmetrically arranged groups of buttons may be different to each other.
(20) In some embodiments, an arrangement of the knobs is symmetrical about the centerline of the control panel.
(21) In some embodiments, the control rings have an upper surface with an outer radius and an inner radius and a surface profile extending in a radial direction that is generally inclined from its outer radius to a point more than half way to its inner radius. The surface profile may be generally inclined from its inner radius to a point less than half way to its outer radius. The control rings may include a substantially cylindrical face around their outer circumference.
(22) The balls of said trackballs may be mounted relative to a top face of the control panel such that the ball has a maximum extension from the top face of the control panel of more than 30% of the diameter of said ball. The extension is measured in a direction parallel with an axis of rotation of the control ring.
(23) The maximum extension may be more than 35% of the diameter of said ball. The maximum extension may be more than 37% of the diameter of said ball. The maximum extension may be more than 40% of the diameter of said ball. The maximum extension may be about 40.5% of the diameter of said ball.
(24) In some embodiments, a control ring may include a coupling that engages with a corresponding coupling mounted in a fixed position with respect to the control panel such that when the control ring is coupled to the corresponding coupling, said control ring retains the ball of a respective trackball in an operating position in the control panel.
(25) The coupling of the control ring may be magnetically attracted to a corresponding coupling of the control panel. The corresponding coupling may include one or more magnets to attract a non-magnetized component of the control ring's coupling.
(26) In some embodiments, supporting a display device of the color correction system in an upright position may comprise supporting the display device of the color correction system such that it is tilted in a distal direction at an angle of less than 40 degrees from vertical. The display device of the color correction system may be tilted in a distal direction at an angle of less than 20 degrees from vertical. The display device of the color correction system may be tilted in a distal direction at an angle of about 17 degrees from vertical.
(27) In some embodiments, a ball of said trackball may extend above a highest point of its respective control ring by more than 15% of the diameter of said ball. The extension is measured in a direction parallel with an axis of rotation of the control ring.
(28) The trackball may extend above a highest point of its respective control ring, by between 23% and 24% of the diameter of said ball.
(29) In some embodiments, the inside diameter of the control ring may be more than 90% of the diameter of the ball of its respective trackball. The inside diameter of the control ring may be less than 100% of the diameter of the ball of said trackball. The inside diameter of the control ring may be about 95% of the diameter of the ball of said trackball.
(30) In some embodiments, the control surface is a battery powered, portable control surface

capable of controlling a color correction system via a wireless communication channel.
(31) In another aspect there is provided a color correction system comprising: a computer system configured to run a color correction application; a display arranged to display a graphical user interface for the color correction application; and a control surface according to an embodiment of the first aspect; the control surface is in data communication with the computer system such that user inputs made using any one or more of said plurality of trackballs, plurality of buttons and plurality of knobs are communicated as control signals to the computer system to control said color correction application.
(32) In some embodiments, the display forms part of a tablet computer. The tablet computer may be supported in the display stand of the control surface.
(33) In some embodiments, the tablet computer may comprise the computer system and display in a single unit.
(34) In some embodiments, the computer system configured to run a color correction application may be separate from the tablet computer and in data communication therewith.
(35) In some embodiments, the computer system comprises a computer server and said tablet computer comprises a client in data communication with said computer server, and wherein a graphical user interface of the color correction application is provided by said client.
(36) In some embodiments, the control surface may be in data communication with the computer system via at least one wireless network connection.
(37) While the invention(s) disclosed herein are amenable to various modifications and alternative forms, specific embodiments are shown by way of example in the drawings and are described in detail. It should be understood, however, that the drawings and detailed description are not intended to limit the invention(s) to the particular form disclosed. Furthermore, all alternative combinations of two or more of the individual features mentioned or evident from the text or drawings comprise additional aspects or inventive disclosures, which may form the subject of claims.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) FIG. **1** shows a control surface according to a first embodiment.
(2) FIG. **2** is a top view of the control surface of FIG. **1** indicating the location of cross sectional views described herein.
(3) FIG. **3** is a side view of the control surface of FIGS. **1** and **2**.
(4) FIG. **4** is a schematic block diagram of the electrical system of the control surface of an embodiment.
(5) FIG. **5** is a cross section along Y-Y in FIG. **2** showing the profile of the display stand of an embodiment.
(6) FIG. **6** is a cross section along Z-Z in FIG. **2** showing the profile of an affordance provided in a display stand of an embodiment.
(7) FIG. **7** is a cross section along X-X in FIG. **2** showing detail of a trackball.
(8) FIG. **8** is a cross section along V-V in FIG. **2** showing, inter alia, detail of a control ring mounting arrangement.
(9) FIG. **9** is a cross section along W-W in FIG. **2** showing, inter alia, detail of a control ring mounting arrangement and encoder.
(10) FIG. **10** illustrates a schematic block diagram of a color correction system according to an embodiment.
(11) FIG. **11** illustrates a color correction system according to an embodiment including a control surface and tablet computer.
(12) FIG. **12** illustrates a schematic block diagram of a color correction system according to an

embodiment which includes a client computer system and server computer system.
(13) FIGS. **13**A to **13**N illustrate several embodiments of control surfaces with different display stands.

DETAILED DESCRIPTION

(14) In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. It will be apparent, however, that the present disclosure may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessary obfuscation of salient details.
(15) FIG. **1** shows a first view of a control surface **100** for controlling a color correction system according to a first exemplary embodiment. The control surface **100**, takes the form of a color grading panel that is able to interface with a computer system (not shown) running a software application that performs color correction functionality.
(16) In general, the control surface **100** includes a control panel **110**. The control panel in this embodiment is generally rectangular in form and has a proximal (front) edge **112** which is nearest a user in normal use, and a distal edge (back) edge (**114**) that is furthest from a user in normal use.
(17) The control panel **110** has a series of controls mounted on it in a position that is accessible to a user, to allow the user to control the color correction system. The control panel **110** comprises the uppermost part of the control surface's **100** housing **300**. The housing **300** may also include a lower enclosure portion **350**. The lower enclosure portion **350** may comprise one or more panels, moldings, chassis, frames, or other members that together define the structure of the housing **300** of the control surface **100**.
(18) The control panel includes a plurality of trackballs **120**. In this example, three trackballs are used (**120**A, **120**B, **120**C). Each trackball (**120**A, **120**B, **120**C) comprises a ball (**122**A, **122**B, **122**C) that interacts with one or more encoders (not shown) to generate a multi-dimensional control signal based on motion of the ball (**122**A, **122**B, **122**C). The ball (**122**A, **122**B, **122**C) of each trackball (**120**A, **120**B, **120**C) is partly exposed above the control panel **110** to enable a user to interact with the ball (**122**A, **122**B, **122**C), and is surrounded by a respective control ring (**124**A, **124**B, **124**C). Each ring (**124**A, **124**B, **124**C) cooperates with at least one encoder (not shown) such that when the control ring (**124**A, **124**B, **124**C) is rotated the rotation is sensed by the respective encoder(s) and a one dimensional control signal can be generated. Each ball (**122**A, **122**B, **122**C) is mounted concentrically with its respecting control ring (**124**A, **124**B, **124**C).
(19) The control panel **110** also includes a plurality of control buttons (**130**). The buttons may be physical buttons as illustrated, or could be buttons on a touch screen interface. In such embodiments the control panel will include a touch screen either in place or in addition to physical buttons. The buttons **130** allow users to make certain inputs to the color correction system in use. The specific input generated by operation of one or more of said button may be fixed, or assignable by the user or color correction system. In some forms, the input generated by operation of one or more of said button can be context sensitive. The buttons **130** on the control panel may be arranged in groups (**132**A, **132**B, **132**C, **132**D, **132**E). The groups can be arranged generally symmetrically about the centerline C-C of the control panel **110**. For example, the middle group **132**E is placed on the centerline C-C, and the arrangement of buttons within that group are symmetrical about the centerline C-C also. The pair of groups **132**C and **132**D are arranged in symmetry with each other about the centerline C-C. Similarly the pair of groups **132**A and **132**B are arranged in symmetry with each other about the centerline C-C. However, although the arrangement of the two groups **132**A and **132**B is symmetrical about the centerline C-C the buttons in the groups **132**A and **132**B are not the same as each other, nor is their arrangement of buttons a mirror images of each other.
(20) The control panel **110** also includes a plurality of knobs **136** coupled to respective rotary encoders. The knobs enable the level of an input to be adjusted in a user friendly and intuitive manner. The knobs may also be arranged symmetrically about the centerline C-C.

(21) The color control surface **100** also includes a display stand **140**. The display stand **140** is able to support a display device of the color correction system in an upright position in use. In this example, the display stand **140** is located adjacent the distal edge **114** of the control panel **110**. The display stand **140** is positioned between the rear-most row of input controls, knobs **136**, and the distal edge **114** of the control panel **110**. But in other embodiments the display stand **140** may be partly, or wholly behind (i.e., further from the user) or aligned with, the distal edge **114** of the control panel **110**.
(22) The display stand **140** in this example, includes a channel **142** extending along the distal edge **114** of the control panel **110**. The channel **142** is sized and shaped to receive an edge of a display device of the color correction system in use. A cross sectional view of the channel **142** of the display stand **140** at position Y-Y is shown in FIG. **5**. FIGS. **5** and **6** also illustrate a lower portion of a display device of the color correction system as it would typically be positioned when supported by the display stand **140** in use. As can be seen in FIG. **5**, the channel **142** has a front support surface **144** that is vertical, or slightly sloped. This front support face **144** limits movement of a supported display device of the color correction system in a proximal (forward) direction. Being slightly sloped in the distal direction (rearward) to form an overhang may minimize the chance of the display device falling backwards and its lower edge sliding up the front support surface. The rear support surface **146** of the channel **142** is sloped backwards and has a radius at its top and bottom. The rear support surface **146** constrains the ability of a supported display device to tilt backwards when it is supported in the display stand **140**. The floor **148** of the channel **142** sits below and supports the display device in use. The radius at the top of the rear support surface **146** provides a lead-in to the channel. A forwardly sloped lead-in segment **144**A is also provided at the top of the front surface **144**. The lead-ins assist a user in inserting their display device in the display stand. By sloping the front lead-in forwards a user will feel a good engagement of a display device with the display stand if the display device is inserted in a substantially vertical orientation. The lower radius that leads to the floor **148** of the channel **142** pushes a supported display device **50** forward so that its front lower corner sits against the front face **144**. The outside ends of the display stand **140** channel **142** may be open or closed. In this case, they are closed to limit sideways movement of the display device.
(23) As illustrated in FIGS. **5** and **6**, the display device **50** of the color correction system will generally be supported in the display stand **140** such that it is tilted back (i.e., in a distal direction) at an angle of more than 10 degrees, but less than 40 degrees from vertical. In this embodiment, when a display device is supported in the display stand **140** it is held an angle of between 15 and 20 degrees from vertical. In particular, in this example it is held at 17 degrees. The exact angle will vary according to the dimensions and shape of the edge of the display device relative to that of the display stand **140**. For example, in the case that a tablet computer or smart phone comprises the computer system of the color correction system, the tablet computer or smart phone itself will also be the display device. For a tablet computer or smart phone with a thickness of approximately 6.5 mm thickness, the display device will rest in the display stand at or close to an angle of 17 degrees to the vertical.
(24) The display stand **140** is also arranged (in size, position, shape and/or layout) with respect to the control panel **110**, so that it can support a display device of the color correction system in a manner which enables the display device's screen to be positioned generally symmetrically about the centerline C-C of the control panel. This may be achieved in some embodiments by making the display stand **140** symmetrical about the centerline C-C of the control panel **110**, as in the present example.
(25) In the present example, the display stand **140** also includes a structure that acts as an affordance **150** to enable access to the display device at a position adjacent to the lowermost edge of the screen. The affordance **150** of the present embodiment takes the form of indentation in the front surface **144** the channel **142**. This causes a localized widening of the channel **142**. The

indentation comprising the affordance **150** is located at the center of the channel **142** and provides physical and visual access to the edge of the display. This is particularly useful if the display device has a special purpose or commonly used interface element located at or near the edge of the display device, such as a home button, front facing camera, sensor, a “swipe up” interface feature such as the “dock” on an iOS device. Such features are more readily accessible and usable via the indentation in the display stand **114**, as the affordance **150** enables a user to insert a finger, stylus or other implement into the locally widened portion of the channel **142** to touch a touch screen or button of the display device, or allows line of sight to a sensor located in that position. In some embodiments, the lateral position of the affordance can be chosen to match a display device of a specific predetermined configuration.

(26) A cross sectional view of the channel **142** at a section along line Z-Z (indicated on FIG. **2**) is shown in FIG. **6**.

(27) At the cross section along line Z-Z, the ordinary shape of the channel's **142** front support surface **144** (shown in dotted lines) has been modified. So, instead of the front face **144** (shown in dotted lines) being substantially vertical or having a small overhang, the front face **151** of the channel **142** at the affordance **150** has a pronounced forward (proximal) slope to increase the width of the entry into the top of the channel. The rear support surface **146** of the channel **142** the same as at cross section Y-Y. The indentation **150** is placed on the centerline C-C **112** of the control panel **110**.

(28) In other embodiments, the display stand can take different forms and include different elements. For example, the stand could comprise any one or more of the following: boss, protrusion, rib, lip, flange, beam, channel, step, shelf, hole, recess, aperture, indentation, slot, cushion, gripping surface, frame, wall, ledge, frame, hook or flap. These structures can perform one or more of the following functions either alone or in combination with another structure: support the weight of the display device of the color correction system in use; constrain any one or more of sliding, twisting, or tilting of the display device of the color correction system in use; act as a fulcrum that supports a display device of the color correction system in use; grip a display device of the color correction system in use.

(29) FIGS. **13**A to **13**L illustrate six alternative examples of display stands that may be used in other embodiments.

(30) In embodiments with a support stand that takes a different form to that of the illustrative embodiment of FIGS. **1** to **12**, the affordance may take a different form also. For example, the affordance may be a gap or space between elements comprising the support structure of the display stand that exposes a similar portion of the screen of the display device.

(31) The display stand **140** of the present embodiment does not protrude above the plane of top face of the control panel **110**. It also provides a straight rear edge for the control panel **110** thus the display stand facilitates the color correction panel **100** having smooth edges. Its location between the rear-most controls (knobs **136**) and the distal edge assists in setting off any protruding elements of the control panel **110** (such as knobs **136**) away from the distal edge of the control panel **110**. This may give the present embodiment advantages when used as a portable device, as there is no additional protrusions or physical structure that can snag or catch when the control surface **100** is stowed or removed from a carrier such as a bag, backpack, sleeve or case.

(32) Many conventional color grading control surfaces have their track balls positioned well back from the proximal edge of the control panel. This facilitates a user positioning their hands over the track balls whilst resting their hands or wrists on top of the control panel during use. In contrast to this, embodiments of the control surface **100** may use any one or more of the following factors: a) a relatively low height of the control panel **110**; b) forward slope of the control panel **110**; and c) proximal positioning of the trackballs **120** to facilitate ease of use of the control surface. FIG. **3** shows a side view of the control surface **100**. As can be seen in side view, the control panel **110** of the control surface **100** has a forward slope from its distal edge **114** to its proximal edge **112**.

(33) Moreover the height of the proximal edge **112** above a surface on which the control surface is resting (e.g., a tabletop) is relatively low. This height may be between 20 mm and 28 mm. In the illustrated embodiment, it may be about 25 mm. The distal edge **114** of the control surface may be between 30 and 50 mm above the resting surface. In some embodiments, it could be between 35 and 40 mm. The illustrative embodiment is about 40 mm high at the distal edge **114** of the control panel and about 38 mm high adjacent to the distal edge of the knobs **136**. The highest point of a track ball's **120** ball **122** may be less than 55 mm, or less than 50 mm above the surface on which the control surface **100** rests in use. In the illustrative embodiment, it can be about 45 mm above the surface. The highest point on the control ring **124** may be less than 45 mm, or less than 40 mm above the surface on which the control surface rests. In the illustrative embodiment, it can be about 37 mm above the surface.
(34) The track balls **120** are positioned in the proximal half of the control panel, i.e., towards the proximal edge **112** of the control surface from the center of the control panel **110**. The center of the trackballs **120** may be less than 70 mm from the proximal edge **112**, and may be less than 65 mm from the proximal edge **112**. In some embodiments, the center of the trackballs may be between 55 and 60 mm from the proximal edge **112**. In the illustrated embodiment, the center of the track ball is about 56 mm from the proximal edge **112**. In some embodiments, the rearmost edge of the control ring **124** of a track ball may be less than 100 mm from the proximal edge **112**, but may be less than 95 mm. In the illustrated embodiment, the rearmost edge of the control ring **124** of a track ball **120** is about 93 mm.
(35) By utilizing a geometry as set out generally above, some embodiments may better preserve a user's experience without retaining the conventional color grading control surfaces' large size. In such embodiments, a user of the control surface **100** may position their hands over the track balls **120** so that they may be used, whilst resting their hands or wrists on a desk/tabletop or other support surface in use in much the same fashion as a conventional computer keyboard would be used (possibly with a wrist support pad if desired).
(36) Such adaptations may also facilitate use of the control surface **100** as a portable device as it may retains overall compact dimensions and usability.
(37) FIG. **4** is a schematic block diagram **400** of the electrical components of the control surface **100**. Overall control is provided by processor **405** which receives inputs from a variety or input sources as discussed below, and generates control signals for outputting on an I/O system.
(38) The circuitry **400** includes, for each trackball **120**A, **120**B, **120**C: at least one, and in some embodiments more than one, encoder **422**A, **422**B, **422**C that senses motion of the ball **122**A, **122**B, **122**C of the track ball **120**A, **120**B, **120**C; and at least one encoder **424**A, **424**B, **424**C to sense rotational motion of the respective ring **124**A, **124**B, **124**C of the trackball **120**A, **120**B, **120**C. The ball encoder(s) **422**A, **422**B, **422**C for each ball **122**A, **122**B, **122**C generate a multi-dimensional control signal based on motion of the ball **122**A, **122**B, **122**C for use in controlling the color correction system. Each encoder may generate a one dimensional output, with its output being combined with that of another encoder to generate a multidimensional control signal for the ball, or alternatively one or more of the encoders can generate a multidimensional output that is used alone (or in combination with other outputs from other encoders) to generate a multidimensional control signal corresponding to motion of the ball. The encoder (or encoders) associated with a control ring may each generate a one dimensional control signal based on the rotational motion of the ring. Multiple encoders may be used to improve resolution or enable averaging of their outputs.
(39) The circuit **400** also includes a plurality of control buttons (or keys) **132**. LED drivers **410** can also optionally be provided to provide illumination of the control panel. For example, buttons may be backlit and may be selectively illuminated or modulated to display their status. Rotary encoders **436** are adjustable via knobs **136** on the control panel **110** and provide a user adjustable input.
(40) The circuit **400** also includes a power supply system **438**. The power supply system **438**

includes a power storage system **446** (e.g., one or more batteries) arranged to provide power for the control surface **100** in use. The power supply system **438** may optionally include a charger **448** for charging the power storage system **446**. Although, in some embodiments, the power storage system **446** may be rechargeable by an external charging system or may be removable or replaceable. The power supply system **438** in this embodiment also includes a power supply **450** which connects the power storage system **446** to the remainder of the circuitry and provides power at the appropriate voltage, if the required voltage is different than the system voltage. For example, the power supply **450** can include one or more regulators (e.g., linear regulator or switched capacitor regulator) and/or switches that run off the system voltage to provide power to different components at the necessary voltage.

(41) In this example, the circuit **400** is provided with a wired interface system **444**. This wired interface **444** can provide power to the battery charger **448** and optionally also a data connection to the color correction system to enable control signals and other data to be exchanged between the control surface and the color correction system. In one embodiment, the wired interface **444** is a USB-C port.

(42) In one embodiment, if the control surface **100** is connected to a source of power via the wired interface **444**, power is drawn preferentially from the wired interface **444**, instead of the power storage system **446**. This can be performed utilizing the charger **448** to direct power to the power supply **450** instead of the power storage system **446**. Any excess power is sent to the power storage system **446** for charging. When wired interface **444** is not connected, power from the power storage system **446** is sent from to the power supply **450** for powering the device as discussed above. As noted above, because the power storage system **446** may accept a removable or replaceable battery, the wired interface system **444** could be omitted from some embodiments.

(43) The interface system **440** includes a wireless communications interface **442**. The wireless communications interface **442** is adapted in use to exchange data and control signals with the color correction system to enable the control surface **100** to control the color correction system. The wireless communications interface **442** may operate according to any wireless communications methodology or protocol but is preferably selected from one of: an IEEE 802.11 wireless standard; Bluetooth, ZigBee or other IEEE 802.15 standard. It may alternatively use free-space optical communication.

(44) Use of a widely accepted wireless communications protocol may facilitate interoperability with different color correction systems. In the illustrative example, the wireless communications interface **442** is a Bluetooth communications interface.

(45) FIG. **7** is a partial cross sectional view along the line X-X indicated in FIG. **2**, through the track ball **120**A showing the control ring **124**A and ball **122**A. As can be seen, the control ring **124**A has an upper surface **200**A. The size of the ring **124**A is generally defined by its outer radius **202** and inner radius **204**. The upper surface **200**A has a radial surface profile that is shown by the cross section in FIG. **7**. In the illustrated embodiment, the surface profile has three primary sections. A substantially cylindrical face **206** that extends around the outer circumference of said control ring. This appears as generally vertical section in the surface profile. The cylindrical face **206** presents a user with a rim or edge with which they can easily manipulate the control ring **124**A. Generally speaking, a user will manipulate the ring using the cylindrical face **206** with the thumb of the hand that is operating the corresponding ball **122**A. Having a cylindrical face **206** and/or an edge on that face provides good tactile feedback to the user. This may be advantageous for a user because they will typically manipulate the controls of the control surface **100**, including the control ring **124**A, **124**B, **124**C while their visual attention is concentrated on the display of the color correction system.

(46) Moving radially inwards from the cylindrical face **206**, the surface profile is inclined up an outer sloped portion **205**A to a point of maximum height **208**. The maximum height may be at a point **208** more than half way along the ring's surface profile to its inner edge **210**. The surface

profile then slopes downward on an inner sloped portion **205**B to its inner edge **210**. In the present embodiment, the control ring has a surface profile that includes almost linear inclined profiles, but in other embodiments the inclines may meet in a wide radius curve or be smoothly curved (e.g., arcuate) from the end of the cylindrical surface to the inner edge **210**. The outer sloped portion **205**A is preferably inclined at an angle that facilitates engagement by a user's fingers in use. Preferably the outer sloped portion, or possibly the entire upper surface **200**A may be textured. In this embodiment, the outer sloped portion **205**A has a pattern of ribs **205**C. The rib pattern can also be seen clearly in FIGS. **1** and **2**. Again as noted above, because a user will typically manipulate control rings **124**A, **124**B, **124**C while their visual attention is concentrated on the display of the color correction system the texture **205**C can provide a tactile feedback to the user and improve grip.

(47) Having a control ring profile that includes an outer cylindrical portion and that then gradually builds in height towards the center of the control ring assist in addressing the dual desiderata of ease of use of the control ring and ease of stowing of a portable device.

(48) Embodiments of the control surface **100** of the present embodiment may be further adapted for use as a portable device by minimizing the vertical height or thickness of the control surface **100**. This also facilitates carrying of the control surface **100** when not in use.

(49) The illustrated embodiment facilitates this by providing balls **122**A, **122**B, **122**C of a relatively small diameter. As will be appreciated, the diameters of the balls **122**A, **122**B, **122**C potentially presents a limitation on the minimum thickness of the control surface **100**. Thus the overall height or thickness of the control surface **100** can be reduced by using smaller balls for the track balls. However the user experience and control accuracy when using small track balls may be impaired.

(50) The balls **122**A, **122**B, **122**C of the present embodiment may have a diameter that is between about 35 mm and 45 mm, but are preferably are about 40 mm in diameter. However although the balls **122**A, **122**B, **122**C are smaller than trackballs used in conventional color correction control surfaces the expected impairment of usability can be mitigated by mounting them relatively high in the control panel **110** and/or relatively high with respect to their respective control ring.

(51) For example, the balls **122**A, **122**B, **122**C of said trackballs are mounted relative to a top face of the control panel **110** such that the ball has a maximum extension (**214**) from the top face of the control panel **110** (indicated by line **215** in FIG. **7**) of more than 30% of the diameter of said ball. The extension of the balls **122**A, **122**B, **122**C above the control panel face may be more than 35%, 37% or 40% of the diameter of the ball **122**A, **122**B, **122**C. In the illustrative embodiment, the maximum extension of the balls **122**A, **122**B, **122**C is about 40.5% of the diameter of said ball. The extension of the trackball can be measured in a direction parallel with an axis of rotation (**220** in FIG. **7**) of the control ring **124**A.

(52) In some embodiments, the ball of said trackball can extend above a highest point of its respective control ring by more than 15% of the diameter of the ball. Again, the extension of the trackball may be measured in a direction parallel with an axis of rotation (**220** in FIG. **7**) of the control ring **124**A. In this example, the line **225** defines a plane that is perpendicular to the axis **220** and is tangential to the highest point **208** on the control ring **124**A. The ball of said trackball may extends above this highest point of its respective control ring by more than 20% of the diameter of the ball. In the illustrative embodiment, the ball of said trackballs extends above the highest point of its respective control ring (dimension **226**) by about 23.5% of the diameter of the ball, and in some embodiments, a greater extension, e.g., above 25%, and up to 40%.

(53) It may also be advantageous to have a relatively large proportion of the ball **122**A exposed through the aperture of the control ring **124**A of the track ball **120**A. That is the inside diameter of the control ring **124**A (i.e., **2***x* the dimension of radius **204** in FIG. **7**) can be relatively large compared to the diameter of the ball. In some embodiments, the inside diameter of the control ring can be more than 90% of the diameter of the ball of said trackball, but less than the diameter of the track ball. The inside diameter of the control ring can be more than 93% of the diameter of the ball

of said trackball. In the illustrative embodiment, the inside diameter of the control ring is about 95% of the diameter of the ball.

(54) This enlarged relative extension of the balls **122**A, **122**B, **122**C through the control ring **124**A, **124**B, **124**C increases the surface area of the ball **122**A, **122**B, **122**C that is exposed to the user for operation of the track ball **120**A, **120**B, **120**C and increases the diameter of the portion of the ball that is exposed for user interaction when seen in plan view. Together these effects may allow control of the trackball in a manner consistent with using a larger trackball on a conventional color control surface. Advantageously the slope of the inner sloped portion **205**B of the control ring (e.g., **124**A) down towards the plane of the control panel **110** enables a user to more readily access the full extent of the exposed surface of the ball (e.g., **122**A) without touching the control ring (**124**A) in the process.

(55) FIG. **7** also shows cross sections though some components of the housing **300** that supports and encloses the internal components of the control surface **100**, and are indicated generally by reference numeral **300**. An encoder **302** in the form of an optical reader can also be seen mounted below the ball **122**A. A plurality of such encoders are used to track the motion of the ball **122**A and generate an output signal for control of the color correction system. The encoder **302** is provided with a cover or lens **304** to protect its optical components from soiling. As will be discussed in more detail in connection with FIGS. **8** and **9**, the control ring **124**A also includes a bearing **306**. The bearing **306** in this example is a set of ball bearings in a race that is interference fit into a recess defined by a downwardly depending flange **308**. The flange **308** extends circumferentially around the underside of the control ring **124**A and the bearing **306** is pushed into the circular rim provided by the flange **308**.

(56) FIGS. **8** and **9** illustrate further details of a trackball **120**A of the control surface **100** of an embodiment. FIG. **8** is a cross section along line V-V in FIG. **2**, whereas FIG. **9** is a cross section along line W-W in FIG. **2**. As will be appreciated, all trackballs in the illustrative embodiment will be the same as each other, and the position of the cross sections is chosen for clarity in FIG. **2**, not to indicate differences between trackballs **120**B and **120**C.

(57) The cross section at V-V shown in FIG. **8** illustrates a cross section through the ball **122**B and control ring **124**B to illustrate the coupling between the control ring **124**B and the control panel **110**. Because there may occasionally be a need to remove and clean the balls **122**A, **122**B, **122**C and encoder sensor covers **304**, it can be advantageous to have removable control rings. Thus the coupling between the control ring **124**B and the remainder of the control panel can advantageously be able to be detached relatively easily. For convenience the coupling is being described as a being between a control ring and the control panel. The actual component or components to which the control ring is coupled may be part of a separable trackball assembly, the control panel chassis, a housing component or other physical structure. However, what is relevant is that control ring is coupled to a structure that held in a fixed relationship to the control panel during use. In the illustrative embodiment, the coupling uses magnetic attraction to hold the components together.

(58) Along cross section V-V the control panel **110** has a magnet **310** mounted in a fixed position. When the control ring **124**B is in its correct operating position the magnet is aligned with the control ring **124**B. Specifically the, bearing race **306** mounted to the control ring **124**B includes or is made of a ferromagnetic material that is attracted to the magnet **310** when in correct alignment. In this example, the inner race **306***i* is correspondingly aligned with the magnet **310** and made of steel, so the control ring **124**B is coupled to the control panel by magnetic attraction. A plurality of such connections can be made at different points around the control ring **124**B. For example, the cross section V-V can be repeated (e.g., regular 90, 120, 180 degree intervals) around the control ring. The flange **308** may be provided with one or more bosses **309** or a ridge placed around its inner circumference to ensure that the bearing **306** remains seated in the rim defined by the flange **308**. In this example, the control ring **124**B includes no magnetized elements that rotate with the control ring **124**B in use. The magnetic elements are provided in a fixed position with respect to the

control panel **110**. This arrangement may minimize induced currents and hence electrical interference in the control circuitry of the control surface **100**.
(59) FIG. **8** also shows ball bearing **315** and support **320** on which the ball **122**B are supported in a manner that provides suitable rolling performance in use. It may also be noted, and possibly as best appreciated in FIG. **7**, that the housing components **300** and general mounting arrangement for the ball **122**A, **122**B, **122**C define a cup **301** in which the ball **122**A, **122**B, **122**C is suspended. However the cup **301** does not fully enclose the bottom of the ball, but instead has an aperture **301**A at its lower side, through with the ball **122**A, **122**B, **122**C protrudes. This aperture avoids the need for an extra wall between the ball **122**A, **122**B, **122**C and the bottom surface of the housing and clearance between the ball **122**A, **122**B, **122**C and the cup **301**, and so assists in minimizing the overall height of the control surface **100**.
(60) FIG. **9** shows a cross section at W-W through the ball **122**C and control ring **124**C to illustrate additional details of the control ring **124**C and the control panel **110**. In particular, FIG. **9** additionally shows an encoder **330**. This may be one of a plurality of such encoders placed to interact with the control ring **124**C. In this example, the encoder **330** is a photointerrupter that detects the position of the flange **308** of the control ring **124**B. The flange **308** can include a profiled lower edge **332**, e.g., a castellated edge having alternating gaps and protrusions, that alternately breaks a beam of light of the photointerrupter or allows it to pass, so that rotational motion of the ring **124**B can be detected and a control signal generated. In one example, the lower edge **332** may have a castellated profile.
(61) FIG. **9** also illustrates an alignment element in the form of a tapered fin **334**, several of which can be placed in a fixed arrangement around the ball **122**C. The fin **334** defines the correct radial alignment of the control ring **124**C with respect to the ball **122**C and coupling magnets. The fin's **334** outer edge defines the correct radial position of the inner edge of the control ring mounting structure (in this case the inner bearing race **306***i*). The top of the fin **334** is tapered to provide a lead in, so that the control ring **124**C can be conveniently aligned during replacement. Instead of individual fins the alignment element could be a full or partial circumferential rim. The cross section W-W can be repeated (e.g., regular 90, 120, 180 degree intervals) around the control ring, although it may not be necessary to repeat the inclusion of additional encoders **330** in each position.
(62) FIG. **10** provides a block diagram that illustrates one example of a color correction system **500** comprising a control surface **100** as described above, and a computer system **1000** running a color correction application. Computer system **1000** includes a bus **1002** or other communication mechanism for communicating information, and a hardware processor **1004** coupled with bus **1002** for processing information. Hardware processor **1004** may be, for example, one or more general-purpose microprocessors, one or more graphics processing unit, or other type of processing unit, or combinations thereof.
(63) Computer system **1000** also includes a main memory **1006**, such as a random access memory (RAM) or other dynamic storage device, coupled to bus **1002** for storing information and instructions to be executed by processor **1004**. Main memory **1006** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **1004**. Such instructions, when stored in non-transitory storage media accessible to processor **1004**, render computer system **1000** into a special-purpose machine that is customized and configured to perform the operations specified in the instructions.
(64) Computer system **1000** may further include a read only memory (ROM) **1008** or other static storage device coupled to bus **1002** for storing static information and instructions for processor **1004**. A storage device **1010**, comprising storage media such as a magnetic disk, SSD or optical disk, may be provided and coupled to bus **1002** for storing information and instructions including the color correction application and/or image and video data that may be subject to color correction.

(65) The computer system **1000** includes a display **1012** (such as one or more LCD, LED, touch screen displays, or other display) that may be coupled via bus **1002** for displaying information to a user of the color correction system **500**. The computer system **1000** may also include an input device **1014**, e.g., a keyboard including alphanumeric and other keys, that may be coupled to the bus **1002** for communicating information and command selections to processor **1004**. The computer system **1000** may also include a cursor control **1016** input device, such as a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to processor **1004** and for controlling cursor movement on display **1012**.
(66) According to at least one embodiment, the techniques herein are performed by computer system **1000** in response to processor **1004** executing one or more sequences of one or more instructions contained in main memory **1006**. Such instructions may be read into main memory **1006** from another storage medium, such as a remote database. Execution of the sequences of instructions contained in main memory **1006** causes processor **1004** to perform the process steps described herein. In alternative embodiments, hard-wired circuitry may be used in place of or in combination with software instructions. The storage medium may also be used to store images or video files on which color correction may be performed.
(67) The terms “storage media” or “storage medium” as used herein refers to any non-transitory media that stores data and/or instructions that cause a machine to operate in a specific fashion. Such storage media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, optical or magnetic disks, such as storage device **1010**. Volatile media includes dynamic memory, such as main memory **1006**. Common forms of storage media include, for example, a floppy disk, hard disk drive, solid state drive (SSD), magnetic tape, or any other magnetic data storage medium, a CD-ROM, any other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, and EPROM, a FLASH-EPROM, NVRAM, any other memory chip or cartridge. Storage media may be local, in the sense it is connected to the bus **1002** or remote, in so far as it is communicatively coupled to the computer system **1000** via network connection. In this regard, computer system **1000** may also include a communication interface **1018** coupled to bus **1002**.
(68) Communication interface **1018** provides a two-way data communication coupling to a communication network **1050** or another device. For example, communication interface **1018** may include an integrated services digital network (ISDN) card, cable modem, ethernet card, satellite modem, USB interface etc. As an example, communication interface **1018** may be a local area network (LAN) card to provide a data communication connection to a compatible LAN. Wireless links may also be implemented. In the present example, the communication interface **1018** includes a wireless interface **1020**. The wireless interface **1020** may operate according to any wireless communications methodology but is preferable selected from one of: an IEEE 802.11 wireless standard; Bluetooth, ZigBee or other IEEE 802.15 standard. It may alternatively use free-space optical communication.
(69) In any such implementation, communication interface **1018** sends and receives electrical, electromagnetic or optical signals that carry digital data streams representing various types of information.
(70) The color correction system **500** also includes a control surface **100**. The control surface **100** includes circuitry **400** as set out in FIG. **4**.
(71) The control surface **100** is communicatively coupled via a network connection **1060** to the computer system **1000**. In particular, the wireless communications interface **442** of the control surface **100** and wireless interface **1020** of the computer system **1000** form a wireless link over which data may be exchanged between them. In this example, the connection **1060** is Bluetooth connection but may be a connection according to a different communications methodology as described herein.
(72) The computer system **1000** is configured to accept user control inputs from the control surface

100 via the wireless link **1060** to control operation of the color correction application running thereon.
(73) FIG. **11** illustrates an exemplary color correction system **1100**, which is generally equivalent to the system **500** of FIG. **10**, comprising, a computer system **1000** in the form of a tablet computer with a touchscreen, and a control surface **100**. The computer system **1000** runs a non-linear editor application that has a color correction capability, such as Davinci Resolve from Blackmagic Design. A color correction graphical user interface **1102** is shown on the screen of display **1012** of the computer system **1000**. Because the computer device **1000** is a tablet computer its display **1012** is integrally formed into the computer system **1000**. In this case, the computer system **1000** and hence its display **1012** are supported in an upright position in the display stand **140** of the control surface **100**. Despite the computer system **1000** and control surface **100** being in physical contact with each other, data communication between them is performed via a wireless communication channel as discussed using Bluetooth. Via the wireless connection with the control surface **100**, user inputs made using any one or more of said plurality of trackballs, plurality of buttons and plurality of knobs, are communicated as control signals to the computer system **1000** to control the color correction application.
(74) For example, in one embodiment, manipulation of the balls **122**A, **122**B, **122**C on the control surface **100** let the user provide inputs to the color correction application to adjust the colors in an image based on lift, gamma and gain tonal ranges respectively. These inputs correspond to moving an indicator point on corresponding two dimensional graphical elements **1104**A, **1104**B and **1104**C of the graphical user interface **1102**. Multiple trackballs may be manipulated simultaneously in some embodiments. The control rings **124**A, **124**B, **124**C may be used to set master levels for each variable.
(75) The control knobs **136** may be configured to allow a user to provide inputs to the color correction application to adjust image parameters, including but not limited to contrast, saturation, hue, temperature, tint, midtone detail, color boost, shadows, highlights.
(76) FIG. **12** illustrates a further embodiment of a color correction system **500**. Features in the present figure that are common with earlier figures will be labelled with common reference numerals, and such features will not be described again for brevity. The control surface **100** is of the type describe in FIGS. **1-9** and may include circuitry **400** as set out in FIG. **4**. Accordingly details of the circuit **400** are omitted for ease of understanding. In overview, this embodiment differs from the previous embodiment in that the display is part of a separate device to the computer system running the color correction application. Such an embodiment may find particular application in a cloud computing situation where the color correction application (or part of its functionality) is executed on a cloud service. In the present embodiment, the display **1012** forms part of a client computer system **1000** in communication with a server computer system **1000**S that runs a color correction application. This arrangement may be advantageous when the client computer system **1000** is a tablet computer, smart phone, smart television or other computing device with relatively low computing power or data storage capability, but may be used in embodiments where such performance constraints are not present.
(77) The local computer system **1000** provides a client application (e.g., a dedicated client application or web browser) which provides a graphical user interface to a color correction application. The control surface **100** is in data communication with a client computer system **1000** using a wireless link **1200** between the wireless communications interface **1020** of the client computer **1000** and the control surface's wireless communication interface **442**. Control signals representing user inputs from the plurality of controls of the control surface **100** are transmitted over the wireless link **1200** to the client computer device **1000**. The client computer system **1000** is also in data communication with a server computer system **1000***s* via second communications connection **1202**. The server computer system **1000**S may be located locally or remotely from the client computer system **1000** so communication between them may be of any suitable form. In

some examples, both the client computer system **1000** and server computer system **1000**S may be connected to a local area network e.g., using wired Ethernet. In other embodiments, they may be connected to each other via the internet, WAN or other network using a suitable network connection mechanism. The connection **1202** may be direct or may not or may not include data transmission via intervening communication network **1050**. Moreover it may include a combination of wired and wireless network connections.

(78) The client computer system **1000** will transmit, inter alia, control signals from the plurality of controls of the control surface **100** over the connection **1202** to the server computer system **1000**S for use in controlling the color correction application. The server computer system **1000**S will transmit inter alia, display data for generating the graphical user interface shown on the display **1012** of the client computer system **1000**. Because the display of the color correction system is remotely located from the server computer system **1000**S, the server computer system **100** may not have a display **1012**S or if it has a display **1012**S the display may be used for tasks other than display of a graphical user interface of the color correction application. In use, the client computer system **1000** will be positioned in the display stand **140** of the control surface **100** so that its display **1012** is visible to a user of the control surface **100**.

(79) FIGS. **13**A to **13**N illustrate seven additional embodiments of a control surface having display stands with alternative features. In these figures, features that are common with earlier embodiments will be labelled with common reference numerals, and such features will not be described again for brevity.

(80) In each case, the display stand **140** is able to support a display device of the color correction system in an upright position in use. In the embodiments of FIGS. **13**E to **13**N, the display stand **140** is positioned between the rear-most row of input controls, knobs **136**, and the distal edge **114** of the control panel **110**. In the embodiments of FIGS. **13**A to **13**D, the display stand **140** is at, or behind the distal edge **114** of the control panel **110**.

(81) In the embodiment of FIGS. **13**A and **13**B, the display stand **140** generally includes a step **1302** that creates a shelf **1304** behind the distal edge **114** of the control panel **110**. The display stand **140** also includes a flap **1306** that is attached to hinges **1308**. The flap **1306** is rotatable between a stowed position **1310** (illustrated in dotted lines in FIG. **13**B) and an operational position **1312** (shown in solid lines in FIGS. **13**A and **13**B). When the flap **1306** is in its operational position, a display of the color correction system can be placed so that it is supported on the shelf **1304** and rests against the flap **1306** at a slightly backwardly angle that is convenient for viewing by a user. The front edge of the display of the color correction system may rest against the front most face **1302**A of the step **1302** to constrain it from sliding forwards. The front face of the step **1302**A is generally aligned with the distal edge **114** of the control panel **110**.

(82) In the embodiment of FIGS. **13**C and **13**D, the display stand **140** generally includes a step **1322** that creates a shelf **1324** behind the distal edge **114** of the control panel **110**. The display stand **140** also includes three protrusions in the form of posts **1326**A, **1326**B, **1326**C positioned along the distal edge of the shelf **1324**. In use, a display of the color correction system can be placed so that it is supported on the shelf **1324** and rests against the flap **1326** at a slightly backwardly angle that is convenient for viewing by a user. The front edge of the display of the color correction system may rest against the front most face **1322**A of the step **1322** to constrain it from sliding forwards. The front face of the step **1322**A is generally aligned with the distal edge **114** of the control panel **110**.

(83) In the embodiment of FIGS. **13**E and **13**F, the display stand **140** generally includes a channel **1332** that extends the full width of the control panel **110**. The channel **1332** has generally vertical side walls and a flat bottom. The front wall of the channel **1332** is provided with a gripping surface **1334**. In use, a display of the color correction system can be placed so that it is supported in the channel **1332** in a similar fashion to the first embodiment. However, rather than having an overhang on the front wall **1336** of the channel **1332** to minimize the chance that the front edge of the display slides up the wall and the display falls backwards, the gripping surface **1334** is provided

to provide extra friction to prevent sliding. The gripping surface **1334** may be provided in several ways, including but not limited to, providing a textured surface on the channel wall **1336**, providing a textured or resilient material, along the channel wall **1336**, providing a rubberized or overmolded finish on channel wall **1336**. The channel **1332** also includes an arcuate widening **1338** at its center to provide an affordance **150** that aids a user in touching the lower part of the display's screen or other interface element.

(84) In the embodiment of FIGS. **13**G and **13**H, the display stand **140** generally includes an aperture **1340** in the form of an elongate slot between the knobs **136** and the distal edge **114** of the control panel **110**. The aperture **1340** extends almost the full width of the control panel **110**. The display stand also includes a shelf **1348** extending from the rear wall of the housing **300**.

(85) In use, a display of the color correction system can be inserted through the aperture **1340** so that it extends down to and is supported on the shelf **1348**. Because the display is captured within the aperture **1340** it cannot fall backwards, and rests against the rear edge **1344** of the aperture at a backwardly tilted angle.

(86) The shelf **1348** may be provided with a cushion surface **1346**. The cushion surface **1346** can be a resilient material, such as rubber strip or rubberized or overmolded surface on the shelf **1348** and can serve to prevent the display from sliding backwards off the shelf **1348**. The front **1342** and rear **1344** sides of the aperture **1340** may be sloped at about the desired support angle for the display to further prevent the display from sliding backwards off the shelf **1348**. The aperture **1340** includes a widened section **1349** at its center to provide an affordance **150** that aids a user in accessing the lower part of the display's screen or other interface element.

(87) FIGS. **131** and **13**J show a further embodiment. In this example, the display stand includes a plurality of flanges **1350**, **1352**, **1354** on the upper surface of the control panel **110** and running parallel to its distal edge **114**. The rear flange **1350** is relatively tall, and the front flanges **1352** and **1354** are somewhat shorter. The front flange **1350** is spaced apart from the front flanges **1352** and **1354** so as to provide a channel **1356** between them to receive a display of the color correction system. In use, a display of the color correction system can be placed so that it is supported on control panel's top surface in the channel **1356** between the flanges **1350** and **1352**, **1354**. The front edge of the display can rest against front flanges **1352**, **1354** to constrain it from moving forwards, while its rear side can lean against the flange **1350** to hold the display at a slightly backwardly tilted angle that is convenient for viewing by a user. The gap **1358** between the two front flanges **1352** and **1354** provides an affordance **150** so that the central part of the lowermost portion of the display is unobstructed.

(88) FIGS. **13**K and **13**L show a further embodiment including a fold away display stand **140**. In this example, the display stand **140** includes a two foldable frames **1360**. Each frame **1360** includes a J-shaped element **1362** that includes a long rear frame leg **1364** and a short front frame leg **1366**. The J-shaped element is mounted on a hinge **1368** and is movable between a stowed position **1372** (illustrated in dotted lines in FIG. **13**K) and an operational position **1370** (shown in solid lines in FIGS. **13**K and **13**L). When the J-shaped elements **1362** is in its operational position, a display of the color correction system can be placed so that it is supported on the between the frame legs **1364**, **1366**. The lower edge of the display sits on the top of the hinge **1368** and is prevented from sliding forwards by the short front frame legs **1366**. The display leans backwards and rests against long rear frame legs **1364** so that it is tilted slightly backwardly angle that is convenient for viewing by a user.

(89) FIGS. **13**M and **13**N show a further embodiment. In this example, the display stand **140** takes the form of a screen rest **1380** that runs parallel to distal edge **114** of the control panel **110**. The screen rest **1380** is shaped to support the display device at a preferred tilt angle without a specific structure to positively locate or hold the display within the screen rest **1380**. The stand includes a back support **1382** that has a front face **1384** angled at the desired tilt angle. The floor **1388** of the screen rest **1380** is angled in the opposite direction to the front face **1384** of the back support **1382**

and meets it at 90 degrees. A display stand can be placed on the floor **1388** of the screen rest **1380** and tilted back against the back support **1382**. In some embodiments, the floor can include a cushion or gripping surface as discussed above to aid in holding the bottom of the display.

(90) The following clauses describe various example embodiments of the present disclosure.

(91) Clause 1. A control surface for controlling a color correction system, the control surface including: a housing including an upwardly facing control panel, said control panel having a proximal edge which is nearest a user in normal use and a distal edge that is furthest from a user in normal use; said control panel including a plurality of controls; the plurality of controls including: a plurality of trackballs, wherein each trackball comprises a ball and a control ring, said ball cooperating with at least one encoder to generate a multi-dimensional control signal based on motion of the ball, said control ring cooperating with at least one encoder to generate a one dimensional control signal based on the rotational motion of the ring, wherein the ball of said trackball is mounted concentrically with said control ring; a plurality of control buttons; a plurality of knobs coupled to respective rotary encoders; a display stand adjacent the distal edge of the control panel for supporting a display device of the color correction system in an upright position in use; a power storage system contained within the housing, said power storage system being arranged to supply power to the control surface in use; and a wireless communications interface over which control signal from the plurality of controls are transmitted for use in controlling the color correction system.

(92) Clause 2. A control surface as set out in clause 1 wherein the power supply system includes one or more of: a battery; and a charging system.

(93) Clause 3. A control surface as set out in either of clauses 1 or 2 the wireless communications interface operates according to one of the following wireless communications methodologies: an IEEE 802.11 wireless standard; Bluetooth, ZigBee or other IEEE 802.15 standard, free-space optical communication.

(94) Clause 4. A control surface as set out in any one of the preceding clauses wherein the display stand comprises any one or more of the following boss, protrusion, rib, lip, flange, beam, channel, step, hole, recess, aperture, indentation, slot, cushion, gripping surface, frame, wall, ledge, shelf, frame, hook and cradle arranged to perform any one or more of the following: support the weight of the display device of the color correction system in use; constrain any one or more of sliding, twisting, or tilting of the display device of the color correction system in use; act as a fulcrum that supports display device of the color correction system in use; grip the display device of the color correction system in use.

(95) Clause 5. A control surface as set out in any one of the preceding clauses wherein the display stand comprises a channel extending adjacent to the distal edge of the control panel, said channel being adapted to receive an edge of a display device of the color correction system in use, said channel having a front support surface that limits movement of a supported display device of the color correction system in a proximal direction, and a rear support surface that constrains tilting of the supported display device of the color correction system in a distal direction.

(96) Clause 6. A control surface as set out in any one of the preceding clauses wherein supporting a display device of the color correction system in an upright position comprises supporting the display device of the color correction system such that it is tilted in a distal direction at an angle of more than 10 degrees from vertical.

(97) Clause 7. A control surface as set out in any one of the preceding clauses wherein the display stand includes an affordance to enable a user to touch a screen of a supported display device of a color correction system, adjacent to the lowermost edge of the screen.

(98) Clause 8. A control surface as set out in clause 7 wherein the affordance enables a user to touch a central part of the lowermost edge of the screen.

(99) Clause 9. A control surface as set out in either of clauses 7 or 8 wherein the affordance is a recess, indentation, groove, or gap.

(100) Clause 10. A control surface as set out in any one of the preceding clauses wherein an arrangement of the trackballs is symmetrical about a centerline of the control panel.
(101) Clause 11. A control surface as set out in clause 10 wherein the display stand can support a display device of the color correction system in a position such that a screen of the display device is symmetrical about said centerline.
(102) Clause 12. A control surface as set out in clause 11 wherein the display stand is symmetrical about said centerline.
(103) Clause 13. A control surface as set out in any one of clauses 10 to 12 wherein the buttons on the control panel are arranged in groups, and wherein said groups of buttons are positioned symmetrically about the centerline of the control panel.
(104) Clause 14. A control surface as set out in clause 13 wherein an arrangement of buttons within at least one pair of symmetrically arranged groups of buttons are different to each other.
(105) Clause 15. A control surface as set out in any one of clauses 10 to 14 wherein an arrangement of the knobs is symmetrical about the centerline of the control panel.
(106) Clause 16. A control surface as set out in any one of the preceding clauses wherein said control rings have an upper surface with an outer radius and an inner radius and a surface profile extending in a radial direction that is generally inclined from its outer radius to a point more than half way to its inner radius.
(107) Clause 17. A control surface as set out in clause 16 wherein said surface profile is generally inclined from its inner radius to a point less than half way to its outer radius.
(108) Clause 18. A control surface as set out in any one of the preceding clauses wherein said control rings include a substantially cylindrical face around their outer circumference.
(109) Clause 19. A control surface as set out in any one of the preceding clauses wherein the balls of said trackballs are mounted relative to a top face of the control panel such that the ball has a maximum extension from the top face of the control panel of more than 30% of the diameter of said ball, wherein said extension is measured in a direction parallel with an axis of rotation of the control ring.
(110) Clause 20. A control surface as set out in clause 19 wherein the maximum extension is more than 35% of the diameter of said ball.
(111) Clause 21. A control surface as set out in clause 19 wherein the maximum extension is more than 37% of the diameter of said ball.
(112) Clause 22. A control surface as set out in clause 19 wherein the maximum extension is more than 40% of the diameter of said ball.
(113) Clause 23. A control surface as set out in clause 19 wherein the maximum extension is about 40.5% of the diameter of said ball.
(114) Clause 24. A control surface as set out in any one of the preceding clauses wherein each control ring includes a coupling that engages with a corresponding coupling mounted in a fixed position with respect to the control panel such that when the control ring is coupled to the corresponding coupling, said control ring retains the ball of a respective trackball in an operating position in the control panel.
(115) Clause 25. A control panel as set out in clause 24 wherein the coupling of the control ring is magnetically attracted to a corresponding coupling of the control panel.
(116) Clause 26. A control panel as set out in clause 25 wherein said corresponding coupling includes one or more magnets to attract a non-magnetized component of the control ring's coupling.
(117) Clause 27. A control surface as set out in any one of the preceding clauses wherein supporting a display device of the color correction system in an upright position comprises supporting the display device of the color correction system such that it is tilted in a distal direction at an angle of less than 40 degrees from vertical.
(118) Clause 28. A control surface as set out in any one of the preceding clauses wherein supporting a display device of the color correction system in an upright position comprises

supporting the display device of the color correction system such that it is tilted in a distal direction at an angle of less than 20 degrees from vertical.
(119) Clause 29. A control surface as set out in any one of the preceding clauses wherein supporting a display device of the color correction system in an upright position comprises supporting the display device of the color correction system such that it is tilted in a distal direction at an angle of about 17 degrees from vertical.
(120) Clause 30. A control surface as set out in any one of the preceding clauses wherein a ball of said trackball extends above a highest point of its respective control ring, by more than 15% of the diameter of said ball, wherein said extension is measured in a direction parallel with an axis of rotation of the control ring.
(121) Clause 31. A control surface as set out in clause 30 wherein ball of said trackball extends above a highest point of its respective control ring, by between 23% and 24% of the diameter of said ball.
(122) Clause 32. A control surface as set out in any one of the preceding clauses wherein the inside diameter of the control ring is more than 90% of the diameter of the ball of said trackball.
(123) Clause 33. A control surface as set out in clause 32 wherein the inside diameter of the control ring is less than 100% of the diameter of the ball of said trackball.
(124) Clause 34. A control surface as set out in clause 32 wherein the inside diameter of the control ring is about 95% of the diameter of the ball of said trackball.
(125) Clause 35. A color correction system comprising: a computer system configured to run a color correction application, a display arranged to display a graphical user interface for the color correction application; a control surface as set out in any one of the preceding clauses; said control surface being in data communication with the computer system such that user inputs made using any one or more of said plurality of trackballs, plurality of buttons and plurality of knobs are communicated as control signals to the computer system to control said color correction application.
(126) Clause 36. The color correction system as set out in clause 35 wherein the display forms part of a tablet computer.
(127) Clause 37. The color correction system as set out in clause 37 wherein the tablet computer is supported in the display stand of the control surface.
(128) Clause 38. The color correction system as set out in either clause 36 or 37 said tablet computer comprises the computer system and display in a single unit.
(129) Clause 39. The color correction system as set out in either clause 36 or 37 said computer system configured to run a color correction application is separate from the tablet computer and in data communication therewith.
(130) Clause 40. The color correction system as set out in 40 wherein said computer system comprises a computer server and said tablet computer comprises a client in data communication with said computer server, and wherein a graphical user interface of the color correction application is provided by said client.
(131) Clause 41. The color correction system as set out in any one of claims 35 to 41 wherein said control surface is in data communication with the computer system via at least one wireless network connection.
(132) Any definitions expressly provided herein for terms contained in the appended claims shall govern the meaning of those terms as used in the claims. No limitation, element, property, feature, advantage or attribute that is not expressly recited in a claim should limit the scope of the claim in any way.
(133) As used herein, the terms “include” and “comprise” (and variations of those terms, such as “including”, “includes”, “comprising”, “comprises”, “comprised” and the like) are intended to be inclusive and are not intended to exclude further features, components, integers or steps.
(134) For aspects of the disclosure that have been described using flowcharts, a given flowchart

step could potentially be performed in various ways and by various devices, systems or system modules. A given flowchart step could be divided into multiple steps and/or multiple flowchart steps could be combined into a single step, unless the contrary is specifically noted as essential. Furthermore, the order of the steps can be changed without departing from the scope of the present disclosure, unless the contrary is specifically noted as essential.
(135) The various embodiments described above can be combined to provide further embodiments. These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled.

Claims

1. A control surface for controlling a color correction system, the control surface including: a portable housing including an upwardly facing control panel, said control panel having a proximal edge which is nearest a user in normal use and a distal edge that is furthest from a user in normal use, the distal edge being substantially straight and having a height of between 30 mm and 50 mm, and said control panel including a plurality of controls, the plurality of controls including: a plurality of trackballs mounted in a proximal half of the control panel, wherein each trackball comprises a ball and a control ring, said ball cooperating with at least one encoder to generate a multi-dimensional control signal based on motion of the ball, and said control ring cooperating with at least one encoder to generate a one-dimensional control signal based on rotational motion of the control ring, wherein the ball of said trackball is mounted concentrically with said control ring; a plurality of control buttons; a plurality of knobs coupled to respective rotary encoders; a display stand adjacent the distal edge of the control panel for supporting a display device of the color correction system in an upright position in use; a power storage system contained within the housing, said power storage system being arranged to supply power to the control surface in use; and a wireless communications interface over which control signals from the plurality of controls are transmitted for use in controlling the color correction system, wherein the display stand comprises a channel extending adjacent to the distal edge of the control panel, said channel being between the distal edge of the control panel and a rearmost control of the plurality of controls, said channel being adapted to removably receive an edge of a display device of the color correction system in use, said channel having a front support surface that limits movement of a supported display device of the color correction system in a proximal direction, and a rear support surface that constrains tilting of the supported display device of the color correction system in a distal direction.
2. The control surface as claimed in claim 1 wherein the display stand further comprises any one or more of the following: a boss, protrusion, rib, lip, flange, beam, step, hole, recess, aperture, indentation, slot, cushion, gripping surface, frame, wall, ledge, shelf, frame, or hook and cradle, arranged to perform any one or more of the following: support the weight of the display device of the color correction system in use; constrain any one or more of sliding, twisting, or tilting of the display device of the color correction system in use; act as a fulcrum that supports display device of the color correction system in use; or grip the display device of the color correction system in use.
3. The control surface as claimed in claim 1 wherein supporting a display device of the color correction system in an upright position comprises supporting the display device of the color correction system such that the display device is tilted in a distal direction at an angle of more than 10 degrees from vertical.
4. The control surface as claimed in claim 1 wherein the display stand includes an affordance to enable a user to touch a screen of a supported display device of a color correction system, adjacent to a lowermost edge of the screen.

5. The control surface as claimed in claim 1 wherein said control rings have an upper surface with an outer radius and an inner radius and a surface profile extending in a radial direction that is generally inclined from the outer radius to a point more than halfway to the inner radius.
6. The control surface as claimed in claim 1 wherein said control rings include a substantially cylindrical face around their outer circumference.
7. The control surface as claimed in claim 1 wherein the balls of said plurality of trackballs are each mounted relative to a top face of the control panel such that the ball has a maximum extension from the top face of the control panel of more than 30% of the diameter of said ball, wherein said maximum extension is measured in a direction parallel with an axis of rotation of the control ring.
8. The control surface as claimed in claim 7 wherein the maximum extension is about 40.5% of the diameter of said ball.
9. The control surface as claimed in claim 1 wherein each control ring includes a coupling that engages with a corresponding coupling mounted in a fixed position with respect to the control panel such that when the control ring is coupled to the corresponding coupling, said control ring retains the ball of a respective trackball in an operating position in the control panel.
10. The control surface as claimed in claim 9 wherein the coupling of the control ring is magnetically attracted to the corresponding coupling of the control panel.
11. The control surface as claimed in claim 1 wherein a ball of said trackball extends above a highest point of its respective control ring by more than 15% of the diameter of said ball, wherein said extension is measured in a direction parallel with an axis of rotation of the respective control ring.
12. The control surface as claimed in claim 1 wherein the inside diameter of the control ring is more than 90% of the diameter of the ball of said trackball.
13. The control surface as claimed in claim 12 wherein the inside diameter of the control ring is less than 100% of the diameter of the ball of said trackball.
14. A color correction system, comprising: a computer system configured to run a color correction application; a display arranged to display a graphical user interface for the color correction application; and a control surface as claimed in claim 1; wherein said display is removably insertable into the channel of the display stand of the control surface, and said control surface is in data communication with the computer system such that user inputs made using any one or more of said plurality of trackballs, said plurality of control buttons, and said plurality of knobs are communicated as control signals to the computer system to control said color correction application.
15. The color correction system as claimed in claim 14 wherein the display forms part of a tablet computer.
16. The color correction system as claimed in claim 15 wherein said tablet computer comprises the computer system and display in a single unit.
17. The color correction system as claimed in claim 15 wherein said computer system configured to run the color correction application is separate from the tablet computer and in data communication therewith.
18. The color correction system as claimed in claim 17 wherein said computer system comprises a computer server and said tablet computer comprises a client in data communication with said computer server, and wherein a graphical user interface of the color correction application is provided by said client.
19. The color correction system as claimed in claim 14 wherein said control surface is in data communication with the computer system via at least one wireless network connection.
20. The control surface as claimed in claim 1 wherein the display stand does not protrude above a plane of the control panel.
21. The control surface as claimed in claim 1 wherein the control panel slopes downwards from the distal edge to the proximal edge.

22. The control surface as claimed in claim 21 wherein the proximal edge of the control panel has a height of between 20 mm and 28 mm.